

Rock Ptarmigan Age and Sex Classification with Machine Learning

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1. Introduction

Researchers in Antarctica are collecting data on Rock Ptarmigan birds to better understand their behavior, population dynamics, and responses to climate change. The problem is to determine a ptarmigan's sex and age using only an image of its wing. This is important because being able to identify these characteristics would speed up data collection and allow researchers to classify birds even while they are in flight. To support this effort, we are designing an algorithm aimed at moving closer to that goal.

2. Literature Review

Alisa Brandt's findings show that in Rock Ptarmigan, age determination is most reliable from wing characteristics, particularly through feather structure, wear, and molt patterns. Juveniles consist of narrower, more worn feathers, with dark pigmentations, while adults display broader, cleaner, and more uniform feathers, with visible molt boundaries acting as temporal indicators of development. These features are subtle and influenced by environmental and individual factors. In contrast, gender differences are weak or not readable from the wings, as the sex is primarily expressed in other regions like the head and body through coloration and markings.

Nielsen tells us that wing length and area differ slightly between sexes: males tend to have longer wings than females. Body size and condition (mass, protein/fat reserves) vary with age: adults are generally more robust, while juveniles are lighter and less developed. Furthermore, external morphology differences (wing shape, tail length, comb/head markings) are subtle but measurable and can act as visual cues for distinguishing age and sex. These findings suggest that age classification can rely on subtle differences in wing structure and body condition, while gender classification depends on localized markers like comb size or head markings.

The study by Kim et al. applies deep learning-based image classification to identify 26 species of Amazon parrots from large image datasets. It utilizes object detection models (specifically Single Shot MultiBox Detector, SSD) with various CNN backbones to detect and classify birds in images. The workflow involved constructing a labeled dataset, using data augmentation to improve model generalization (horizontal flips, rotations, zooms), and training an object detection network to capture the bird body. This work shows that deep learning and object detection can learn subtle visual features from images to classify biological categories, which parallels our task of distinguishing Rock Ptarmigan age and sex from wing photos.

The paper by Saaketh introduces Bilinear CNNs (B-CNNs), which capture subtle texture and pattern differences by modeling pairwise feature interactions, making them effective for fine-grained image classification. Applied to datasets like Caltech-UCSD Birds 200, the model performs well on visually similar bird categories with small, localized differences. This is relevant

to our project since wing-based identification depends on fine feather details (e.g., wear and pigmentation). It suggests that using fine-grained models like B-CNNs can improve detection of subtle features for age classification, and potentially capture weak structural cues related to sex.

3. Methods

Our initial approach used rule-based color thresholding to estimate age by measuring the proportion of dark pixels in each image. Images with more black pigmentation were classified as juveniles, while those with less were classified as adults. Although simple and fast, this method could miss subtle patterns, and the black image backgrounds also interfered with detection.

We then shifted to a neural network approach inspired by image classification work previously completed with the MNIST dataset. Two different approaches were explored for implementing the multilayer perceptron (MLP). The first approach involved manually coding the MLP architecture, while the second used the PyTorch framework to implement the same model. Although the PyTorch implementation achieved higher accuracy for sex prediction, it produced lower accuracy for age prediction. Because the manually coded MLP produced more balanced and consistent performance across both prediction tasks, it was selected as the final approach used in this project. We standardized image names into an `img_name_year` format to avoid duplicate conflicts. Metadata was extracted using ExifTool, and only the relevant fields (`age_label`, `sex_label`, and `image_name`) were retained. Because metadata formatting was inconsistent, regular expressions were used to correctly parse values before exporting the cleaned labels into a CSV file.

Initially, the network was trained to jointly predict both age and sex, but the model struggled to accurately learn the joint probability distribution. As a result, we separated the task into two independent classifiers, which produced better performance. We experimented with models containing 20, 50, and 100 hidden nodes. We also experimented with two different batch sizes, 32 and 64. Although a batch size of 32 took significantly longer to process, it delivered better results. During training, we also observed that the model began overfitting after approximately 42 epochs, so the final training cycle was limited to 42 epochs.

4. Results

The joint classifier achieved only 29.23% validation accuracy, showing that the model struggled to learn the combined probability distribution of age and sex simultaneously. Because of this poor performance, the task was separated into two independent classifiers for age and sex prediction. This significantly improved the results, with age classification reaching 61.54% validation accuracy using 20 hidden nodes with a learning rate of 0.01, and sex classification reaching 76.15% validation accuracy using 50 hidden nodes with a learning rate of 0.01. These results indicate that a smaller model was more effective for age prediction, whereas sex prediction improved when using a larger hidden layer. Table 1 summarizes the final accuracies across the different model configurations. To better interpret individual predictions, a Softmax activation function was applied to produce class probabilities, with the resulting outputs shown in Tables 2 and 3. For example, the first age prediction from the test set produced a 35%

probability that the bird was classified as an adult and a 65% probability that it was classified as a juvenile. Since juvenile had a higher probability, the model classified the bird as juvenile, which matched the true label and therefore represented a correct prediction. Overall, the model remained relatively consistent for single-variable predictions across node sizes, but accuracy declined sharply when switching to a two-variable setting.

Table 1: Final Accuracies

hidden nodes	age test	sex test
20	0.6154	0.7308
50	0.6077	0.7615
100	0.5462	0.6923

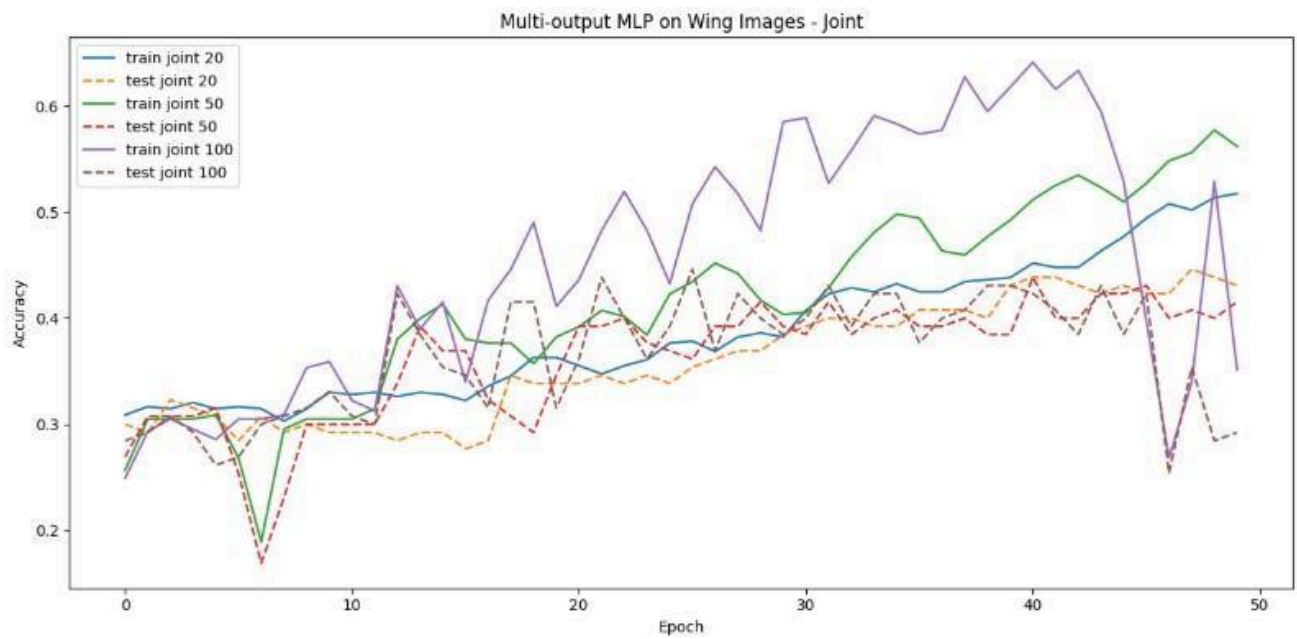
Table 2: Softmax algorithm age results (confidence scores)

Probability of adult (first 5 samples)	Probability of juvenile (first 5 samples)
35%	65%
40.3%	59.7%
42.5%	57.5%
36.4%	63.6%
34.7%	65.3%

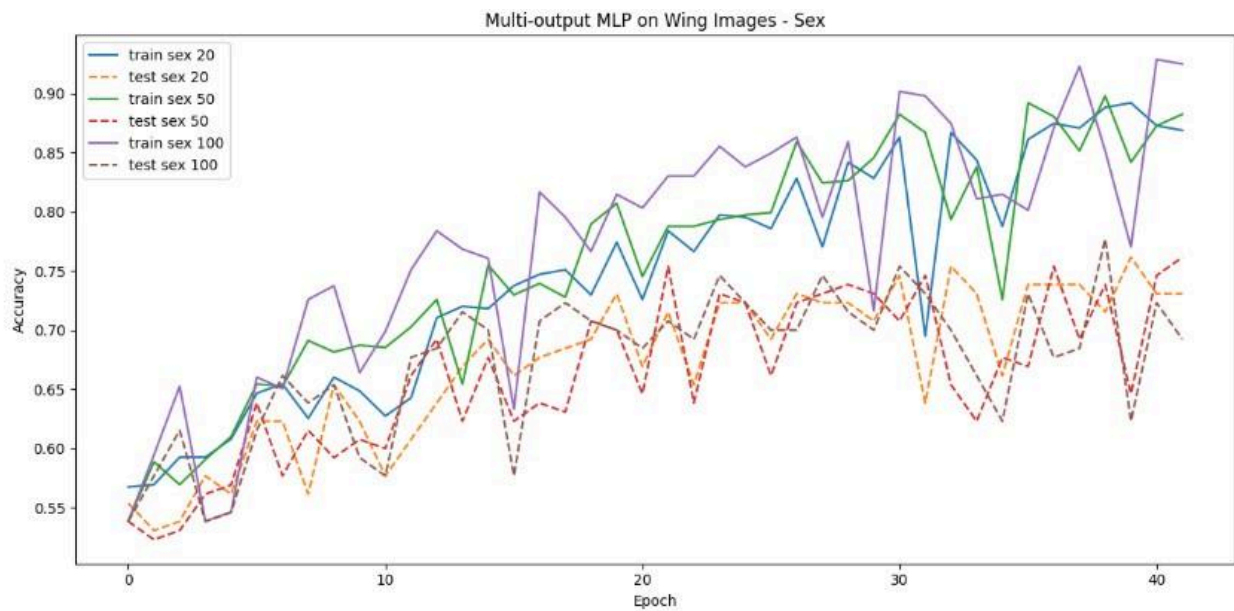
Table 3: Softmax algorithm sex results (confidence scores)

Probability of Male (first 5 samples)	Probability of Female (first 5 samples)
43.9%	56.1%
48.9%	51.1%
45.3%	54.7%
45.3%	54.7%
44.8%	55.2%

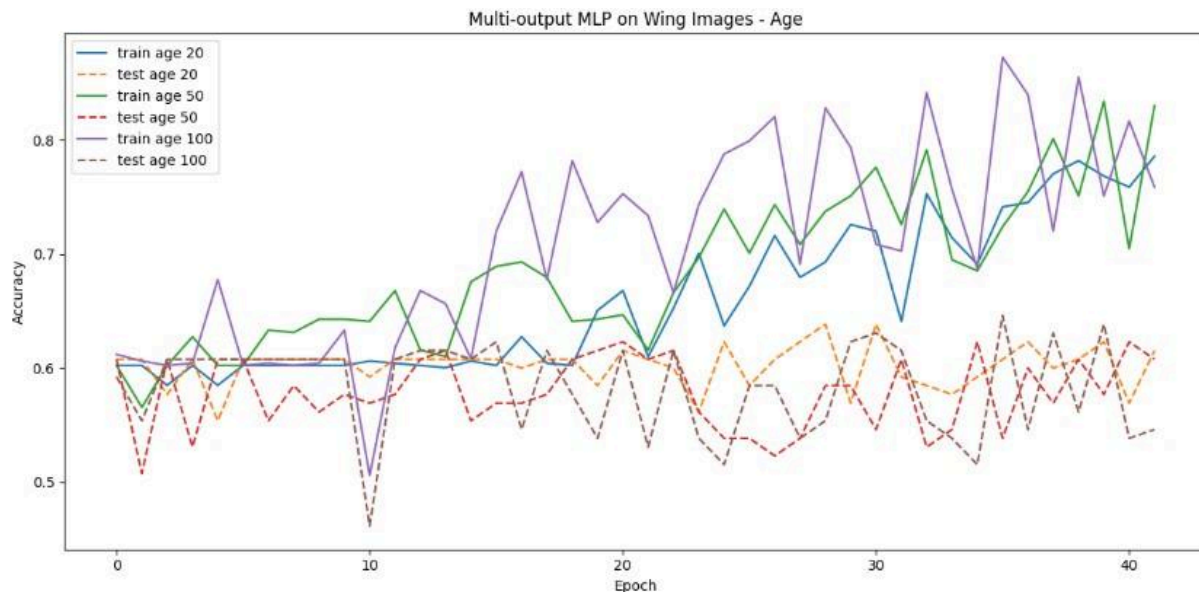
Graph 1: MLP accuracy of the joint predictor of age and sex.



Graph 2: MLP accuracy of the sex predictor



Graph 3: MLP accuracy for the age predictor



The key detail to focus on in the graphs above is the y-axis. For single-variable predictions, training set accuracy reaches around 80-90% as seen in graphs 2 and 3, whereas the joint classifier's max is 65% as seen in graph 1. This supports the observation that predicting age and sex simultaneously made the learning task significantly more difficult for the network. Another important trend from Graph 1 is that the model begins to overfit after approximately 42 epochs. Based on this, 42 epochs was selected as the optimal stopping point, since performance begins to decline beyond this stage.

5. Conclusion

We expect our models to continue showing a clear difference between age and sex prediction. The early experiments already indicate that the classifier is more reliable when the tasks are separated, as seen in the accuracy reaching around 61.54% and 76.15% for age and sex, respectively, with both models improving steadily. However, when these two tasks are combined into a joint classifier, the accuracy drops sharply, showing that the model struggled to learn the combined probability distribution of age and sex simultaneously.

Although we initially expected age prediction to outperform sex prediction based on biological data, the final results showed the opposite. While the wings of the Rock Ptarmigan only give a weak sex differentiation, the model may be leveraging subtle statistical patterns within the dataset that are not immediately apparent by biological interpretation alone. These findings suggest that more advanced models like CNNs may be necessary to extract the detailed features required for reliable age classification while sex prediction appears to benefit more from model complexity than initially anticipated.

6. Contributions

Jay Mhaikar will focus on coding the algorithms, designing the model and incorporating it with the training dataset. Minh Thuong Nguyen will log project reports, note down the results of the search algorithm and other related details, and run test cases. Khoa Bui will help format all report logs, code implementation section, and test results into presentable documentation, with the big focus on organizing all important data. Each member will be helping in the final presentation as follows: Thuong will open the presentation by giving the introduction and discussing the literature related to the project, then Jay will focus on explaining the method, classifier algorithm, discussing the results and findings, and Khoa will conclude the presentation and discuss how the project may further help the image classification of rock ptarmigan.

Works Cited

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